

Shao-Heng Ko | 柯劭珩

Department of Computer Science – Duke University

✉ shaoheng.ko@duke.edu • 🌐 Website

Education

Duke University <i>Ph.D., Computer Science, Certificate in College Teaching program</i> Dissertation (tentative): Computing students' help-seeking approaches and behavior throughout the curriculum: 2021-2025	2020-2026(est.) Advisor: <i>Kristin Stephens-Martinez</i>
National Taiwan University <i>M.S., Graduate Institute of Electrical Engineering</i> Thesis: Encouraging Peer Grading in Massive Open Online Courses	2015-2017 Advisor: <i>Ho-Lin Chen</i>
National Taiwan University <i>B.S., Electrical Engineering</i>	2011-2015

Professional Experience

Duke University <i>Instructor of Record</i>	2024
Institute of Information Science, Academia Sinica <i>Full-time Research Assistant (Research area: approximation algorithms and social network)</i>	2017-2020
Lab of Teaching Innovation, National Taiwan University <i>Massive Open Online Courses (MOOC) Explorer, <i>Manufacturer</i>, and <i>Promoter</i></i>	2015-2017

Honors & Awards

ACM ITiCSE Outstanding Reviewer	2025
Duke Graduate School Dean's Award for Excellence in Teaching	2025
Duke Graduate School Bass Instructor of Record Fellowship	2024
Duke CS Outstanding Teaching Award	2023
Duke CS Outstanding Teaching Award	2021
NTU GIEE Best Master Thesis (Title: Encouraging Peer Grading in Massive Open Online Courses)	2017

Publications

Asterisks (*) indicate equal contribution. Alpha (α) symbols indicate theoretical papers that followed the alphabetical order convention.
Bolded names indicate the person(s) who gave the conference talk.

Conference Proceedings (Full Research Papers)

- [1] **Shao-Heng Ko** and Kristin Stephens-Martinez. Connecting Computing Students' External Help Resource Preferences and Internal Help Resource Usage: 2021-2025. In *ACM SIGCSE TS (forthcoming)*, 2026.
- [2] **Shao-Heng Ko**, Matthew Zahn, Kristin Stephens-Martinez, Yesenia Velasco, Lina Batestilli, and Sarah Heckman. Relationships Between Computing Students' Characteristics, Help-Seeking Approaches, and Help-Seeking Behavior in Introductory Courses and Beyond. In *ACM ICER*, pages 313–326, 2025.
- [3] **Shao-Heng Ko** and Kristin Stephens-Martinez. Prior What Experience? The Relationship Between Prior Experience and Student Help-Seeking Beyond CS1. In *ACM ITiCSE*, pages 100–106, 2025.
- [4] **Shao-Heng Ko**, Kristin Stephens-Martinez, Matthew Zahn, Yesenia Velasco, Lina Batestilli, and Sarah Heckman. Student Perceptions of the Help Resource Landscape. In *ACM SIGCSE TS*, pages 596–602, 2025.
- [5] **Shao-Heng Ko** and Kristin Stephens-Martinez. The Trees in the Forest: Characterizing Computing Students' Individual Help-Seeking Approaches. In *ACM ICER*, pages 343–358, 2024.
- [6] **Shao-Heng Ko** and Kristin Stephens-Martinez. What Drives Students to Office Hours: Individual Differences and Similarities. In *ACM SIGCSE TS*, pages 959–965, 2023.
- [7] Shao-Heng Ko*, **Erin Taylor***, Pankaj K. Agarwal, and Kamesh Munagala. All Politics is Local: Redistricting via Local Fairness. In *NeurIPS*, pages 17443–17455, 2022.
- [8] Pankaj K. Agarwal α , Shao-Heng Ko α , Kamesh Munagala α , and **Erin Taylor** α . Locally Fair Partitioning. In *AAAI*, pages 4752–4759, 2022.
- [9] **Shao-Heng Ko** α and Kamesh Munagala α . Optimal Price Discrimination for Randomized Mechanisms. In *ACM EC*, pages 477–496, 2022.
- [10] **Shao-Heng Ko**, Ying-Chun Lin, Hsu-Chao Lai, Wang-Chien Lee, and De-Nian Yang. On VR Spatial Query for Dual Entangled Worlds. In *ACM CIKM*, pages 9–18, 2019.

Conference Proceedings (Full Experience Reports)

[1] Shao-Heng Ko, Alex Chao, and Violet Pang. Satisfactory for all: supporting mastery learning with human-in-the-loop assessments in a discrete math course. In *ACM SIGCSE TS*, pages 589–595, 2025.

Journal Articles

- [1] Janet Jiang, Jonathan Liu, Shao-Heng Ko, Kristin Stephens-Martinez, and Diana Franklin. Adapting an ecological belonging intervention to an introductory computer science course. *Under submission at ACM Transactions on Computing Education (TOCE)*.
- [2] Janet Jiang, Shao-Heng Ko, and Kristin Stephens-Martinez. Peer instruction in hybrid computing courses: The relationships between student modality, discussion, and learning. *Under submission at ACM Transactions on Computing Education (TOCE)*.
- [3] Shao-Heng Ko and Kristin Stephens-Martinez. Rethinking computing students' help resource utilization through sequentiality. *ACM Transactions on Computing Education (TOCE)*, 25(1), 2025.
- [4] Shao-Heng Ko^α and Kamesh Munagala^α. Optimal Price Discrimination for Randomized Mechanisms. *ACM Transactions on Economics and Computation (TEAC)*, 12(2), 2024.
- [5] Chih-Ya Shen*, Shao-Heng Ko*, Guang-Siang Lee, Wang-Chien Lee, and De-Nian Yang. Density Personalized Group Query. *The International Journal on Very Large Data Bases (VLDB)*, 16(4):615–628, 2022.
- [6] Shao-Heng Ko, Hsu-Chao Lai, Hong-Han Shuai, Wang-Chien Lee, Philip S. Yu, and De-Nian Yang. Optimizing Item and Subgroup Configurations for Social-Aware VR Shopping. *The International Journal on Very Large Data Bases (VLDB)*, 13(8):1275–1289, 2020.

Other Peer-Reviewed Publications

- [1] Matthew Forshaw, Cristina Adriana Alexandru, Caitlin Bentley, Vladimiro González-Zelaya, Joseph Kwame Adjei, Vangel Ajanovski, Mireilla Bikanga Ada, Julian Brooks, Joshua Burrridge, Alex Chao, Rutwa Engineer, Olga Glebova, Tasmina Islam, Mitsuka Kiyohara, Shao-Heng Ko, Ellert Smári Kristbergsson, Svetlana Peltsverger, Seán Russell, Maíra Marques Samary, Merel Steenbergen, and Carolin Wortmann. Fairness in student allocation and group formation (working group abstract; report under submission). In *ACM ITiCSE-WG*, pages 699–700, 2025.
- [2] Salma El Otmani, Janet Jiang, Shao-Heng Ko, and Kristin Stephens-Martinez. The Relationships Between Modality, Peer Instruction Discussion, and Class Sentiment in Hybrid Courses (Poster). In *ACM SIGCSE TS*, pages 1634–1635, 2024.
- [3] Shao-Heng Ko. Characterizing Computing Students' Academic Help-seeking Behavior (Doctoral Consortium Extended Abstract). In *ACM ICER*, pages 73–75, 2023.

Teaching Experiences

See my [website](#) for further descriptions of my roles and responsibilities.

Instructor of Record, Duke CS

- CS230 Discrete Mathematics for Computer Science [Spring 2024 (138 students, 3 graduate TAs, 21 UTAs)] [Exp. Report] [Blog Series]

Teaching Assistant, Duke CS

- CS171 Learning How to Learn with AI Fall 2025 (24)
- CS230 Discrete Mathematics for Computer Science Summer 2025 (8)
- CS330 Intro to the Design and Analysis of Algorithms Spring 2025 (336)
- CS230 Discrete Mathematics for Computer Science Fall 2023 (121)
- CS216 Everything Data Spring 2023 (234)
- CS216 Everything Data Fall 2022 (208)
- CS330 Intro to the Design and Analysis of Algorithms Fall 2021 (142)
- CS230 Discrete Mathematics for Computer Science Spring 2021 (120)
- CS330 Intro to the Design and Analysis of Algorithms Fall 2020 (172)

Teaching Assistant, NTU EE/GIEE

- EE5182 Advanced Algorithms Spring 2017 (97)
- EE5048 The Design and Analysis of Algorithms Fall 2016 (157)
- EE2008 Discrete Mathematics Spring 2016 (169)
- EE5048 The Design and Analysis of Algorithms Fall 2015 (152)

Academic Services

Conference Reviewing

- ACM Conference on International Computing Education Research (ICER) 2026, 2025
- ACM Technical Symposium on Computer Science Education (SIGCSE TS) 2025, 2024, 2023
- ACM Conference on Innovation and Technology in Computer Science Education (ITiCSE) 2025 (*Outstanding Reviewer), 2024
- ACM Designing Interactive Systems Conference (DIS) 2025
- ACM Virtual Global Computing Education Conference (SIGCSE Virtual) 2024
- ACM The Web Conference (WWW) 2024
- IEEE Global Communications Conference (GLOBECOM) 2018

Journal Reviewing

- ACM Transactions on Computing Education (TOCE) 2024-now

Research Mentoring

Undergraduate Students (Duke)

◦ Alex Pool, Optional groupwork in CS theory course assignments	Fall 2025
◦ Ricardo Urena, Relationships between student learning beliefs and help-seeking in CS	Summer 2025
◦ Zehavi Rodriguez, Demographic factors and self-assessment trajectories in intro CS	Summer 2025
◦ Janet Jiang, Effectiveness of Peer Instruction in Hybrid Courses	Summer 2023 - Spring 2025
◦ Salma El Otmani, Effectiveness of Hybrid Courses	Summer 2023
◦ Jerry He, Effectiveness of Hybrid Courses	Summer 2023
◦ Belle (Hao) Xu, Student Behavior on Formative Assessments and Performance on Summative Assessments	Spring 2023
◦ Rhea Tejwani, Efficacy of Office Hours in CS1	Spring 2023

M.S. Students (Academia Sinica-NTU)

◦ Ta-Che Hsiao	2019-2020
◦ Chi-Jen Lo	2019-2020
◦ Chiao-Wen Lin	2019-2020

Talks and Media Appearances

Please see the Publication Section for conference paper presentations.

Podcasts

◦ Guest on <i>The CS-Ed Podcast</i> , S4E13: "Academic Help-Seeking Approaches and Behavior"	11/3/2025
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Panels

◦ "Discipline-based research writing", at <i>Duke Thompson Writing Program 315S Argument Across the Disciplines</i>	10/3/2024
◦ "What Ph.D. programs are like", at <i>Duke Plus Programs</i>	7/5/2023
◦ "Why MOOCs?", at <i>NTU Teaching Innovation Lab</i>	12/26/2016

Invited Research Talks

◦ "Characterizing computing students' help-seeking behavior in office hours, now, in our own classrooms", at PSUT CSEd Group	5/29/2023
◦ "Characterizing computing students' academic help-seeking behavior across courses and help resources", at UCSD CSEd Lab	5/15/2023

Guest Lecturing

◦ Probability in the Real World (2), at Duke CS171 Learning How to Learn with AI	10/30/2025
◦ Probability in the Real World (1), at Duke CS171 Learning How to Learn with AI	10/28/2025
◦ Introduction to Probability, at Duke CS171 Learning How to Learn with AI	10/21/2025
◦ Exam Review, at Duke CS230 Discrete Mathematics	6/18/2025
◦ Random Variables and Expectation, at Duke CS230 Discrete Mathematics	6/17/2025
◦ Probability, at Duke CS230 Discrete Mathematics	6/16/2025
◦ Permutations and Combinations, at Duke CS230 Discrete Mathematics	6/12/2025
◦ Introduction to Counting, at Duke CS230 Discrete Mathematics	6/11/2025
◦ NumPy and Pandas, at Duke CS216 Everything Data	9/11/2024
◦ Counting and Probability, at Duke CS230 Discrete Mathematics	10/18/2023
◦ Proofs by Induction, at Duke CS230 Discrete Mathematics	9/13/2023
◦ Visualization (2), at Duke CS216 Everything Data	3/31/2023
◦ Visualization (1), at Duke CS216 Everything Data	3/29/2023

Miscellaneous

2014: Co-editor of *Benson's amazement in probability*, a collection of student-generated peer assessments in flipped undergraduate probability classes in Taiwan. ISBN: 9789861371832